



Sameto AG Drive Pvt. Ltd.

# Powder Transfer System & Material Handling System (**PTS**)



# Powder Transfer System

Powder Conveying systems has become one of the most economical and effective methods of transfer of material. In a granulation line, all the individual process equipments like Granulator, Fluid Bed Driers Processors, Milling and Sifting machines and most importantly Blenders, need to be charged, discharged and subsequently conveyed to the successive machines in line. All the above conveying operation can be done through the powder transfer systems. Based upon our experience, we have extended our conveying expertise to various allied industries namely Chemical/API and also the food industries.

## features

- Reliable low weight System
- Easy Installation & control
- Gentle to the Material
- Lower Maintenance & Easily cleaned
- MQC-Stainless Steel Clad MS column / SS column.
- Dust free Material Transfer
- Closed column construction with floor and ceiling fastening, pivoted execution
- Almost no wear in entire process
- Design is cGMP – Current Good Manufacturing Practices compliance.
- All contact parts AISI 316 & non contact parts AISI 304.
- Capacity available 100 Kg to 5000 Kg/hr.
- Can be used for transfer of products having bulk densities up to 1.0gm/cc.
- Total Discharge no Retention.
- No segregation or degradation of Product.
- No tool required for opening & cleaning.
- Fully automatic system, with timing controls & automatic air blow back system for filter cleaning.
- Vacuum Blower and cyclone filter can be installed outside production area.
- Avoids product waste, loss and contamination
- Easy to clean and wash of contact parts.



## applications

The Powder Transfer System has many advantages over conventional conveying systems like screw conveyor or bucket elevators. We have supplied this system to a number of clients in the following industries:

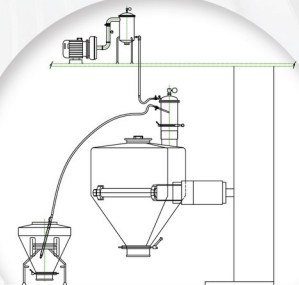
- Chemicals
- Food Industry
- Pharmaceutical Industry
- Animal Feed Industry
- Fertilizer Industry
- Plastics
- Metal Powders
- Colour and Coatings

## Intermittent Vacuum Loading

In the case of indirect loading, the PTS is mounted directly above the blender. Material is conveyed directly from the hoppers or Bin or cylindrical containers through a special suction bayonet. The material is sucked and transferred to the PTS mounted above the machine. The PTS acts as a holding vessel and discharges the material through a fill / dump principle.

The advantage in this application is that the blender body (or any other receiving body) need not be vacuumized.

There has to be an additional height of at least 850mm above the machine which has to be connected with the PTS. The receiver can be mounted on a lifting and lowering device (ideally used so that the filter can be cleaned and refitted) or it can be fixed on a stationary pillar.

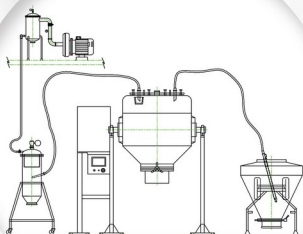


## Continuous Vacuum Loading

The blender in this case is used as a receiving vessel. It is important to note that this method is applicable to any mixer, reactor or vessel capable of withstanding vacuum.

In continuous loading, material is conveyed to the blender and allowed to fall out into the blender vessel. A small filter receiver is connected to the exhaust or vent port of the blender in order to eliminate carryover /dust from the blender. This PTS is equipped with a small compact filter which is periodically pulsed so that any residual dust on the filter is sent back to the blender.

The continuous loading design ensures that all the material is conveyed directly to the blender with minimum spills or waste. Also the room height can be minimal because the PTS is floor mounted and portable.



# How PTS works

step1

The vacuum pump generates a vacuum in the PTS tank. Air rushes in through the hose/pipe and the suction inlet from the feeding point. The bulk material is aspirated and then carried in this air stream.



step2

Inside the PTS tank a filter module separates air and product. The transported material is kept and collected in the tank.

step3

When the PTS is filled with product, the vacuum pump switches off, inside of the vacuum conveyor the pressure is balanced to surrounding area within tenths of a second. The separated material discharges from the vacuum conveyor through the discharge element and falls directly into the unit or tank to be charged.

step4

During discharge of material, the filter module is cleaned automatically by an air purging system. The occasional filter cake is released from the filter unit. CPEC offers piston vibrators and fluidizing units to improve the discharging for sticky bulk materials.

step5

After unloading the product the discharge element closes and the whole conveying cycle starts again

## accessories

### Feed Point & Pipes Feed Station

To manually feed the conveyor direct from bags, barrels, buckets, etc. The air flow intake into the nozzle can be adjusted to optimize the transport of the product into the conveying line.

The feed station should be placed under a big bag or for dumping small bags. It has fluidization as an option for sticky, bridging, non-free flowing products to make the transport problem-free.

### Feed Adapters

The feed adapters are suitable for Chemiplant's feed station, customized hoppers, big bag un-loaders or silos. The feed adapters can adjust the carrying air to optimize the transport of the product into the conveying line.

### Piping

Pipes have lower friction (faster & gentler conveying) than hoses & should be used in all stationary installations. Total conveying distance, vertical distance, number of pipe bends, diameter, whether our customer are using steel tubes, vacuum hoses or a combination of pipes & hoses, the type of connections between the pipes, these are all factors to consider when you choose piping system.

## Options / Variants

- 304 stainless steel or 316 stainless steel construction
- Slide gate, dump gate, rotary valve, or butterfly valve to discharge your material
- Portable base increases installation flexibility
- Standard, food grade or sanitary finishes available
- Teflon coating, air pads, or electro polish for difficult flowing materials

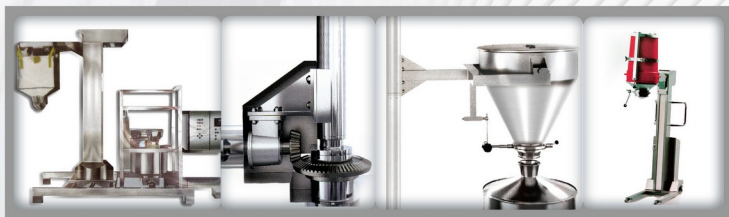
# Lifting & Tilting Device

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Chemiplant's Lifting & Tilting Device are manufactured and projected to support Pharma Bins, Powder Transfer System and Drum Blenders. They are mainly used to optimize the production processes for the handling of powder (handling, dosing, mixing) and avoid the 'cross-contamination in the chemical, pharmaceutical and food industries Upon request it can be manufactured also according to the Atex Directive.

All functions of the Lifting & Tilting Device (Both vertical and horizontal movement as well as all customizable technical specifications) are automatically managed through the control panel. The Lifting & Tilting Device could be Customized and used for multiple functions and/or applications such as:

- Lifting & Tilting Device for pharma bins (feeding)
- Lifting & Tilting Device for drums (feeding)
- Lifting & Tilting Device for Blenders (mixing)



## Hydraulic Lifter

Chemiplant's Hydraulic Lifter are available in different options with unique and modern technologies to optimize operations in the pharmaceutical and food processing industries.

Chemiplant's Hydraulic Lifter features a unique technology such as a vacuum feeding system, chassis trolley, inversion system, telescopic post, hydraulic system & lifting drum.

The Chemiplant's hydraulic lifter design is such that you can use it with other pharmaceutical equipment such as blenders, coater, tablet presses, capsule filling machines, etc.

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# Lifting & Tilting Device

## working principle

The Lifting & Tilting Device is composed of chassis, column and lifting/lowering system & inversion system. For material loading / unloading first lock the FBD Bowl with the conical adapter, and then press "Raise" button to raise the bowl. When it's raised in position, invert it 180° then release the brake and slew the column to swivel the bowl to the process required position. Lower the bowl to the working height and open the butterfly valve to discharge the materials into the next processing equipment.

## application

This equipment is mainly used in the solid dosage production area for transferring and discharging materials. Consequently it is operated with corollary equipments such as granulator, blender, tablets press, capsules filling machine, etc... Pharma Lifters are widely used in pharmaceutical, chemical, foodstuff industries. etc..

## highlight features

- Top & bottom supports provide Completely enclosed structure
- Silent & precise Lifting & Positioning
- Operation is dust & Leak free
- High quality, low friction seals of imported make
- Motion control valve provided for tilting operations
- Load holding valve mounted on the cylinder to ensure complete safety
- Lead screw, Chain & sprocket type designs also available.
- Hard to discharge materials can be easily transported
- Widely used in the Pharmaceutical, Chemicals, Cosmetics, Agrochemicals, Food, Nutraceutical & Animal Feed.



# Hydraulic Lifter

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## application

This equipment is mainly used in the solid dosage production area for transferring and discharging materials, consequently it is operated with corollary equipments such as granulator, blender, tablets press, capsules filling machine Pharma Lifters are widely used for holding PTS, for Blending (mixers), for drum feeding & stacking.



## highlight features

- No dead angles and protruding screws
- Equal distribution of load due to Hydraulic cylinder connected to the lift arm having two rods
- Easy to operate and maintain
- Efficient dust handling design
- Design meets the CE, FDA and cGMP requirements
- Ensures safe operation
- Closed column construction with floor and ceiling fastening, pivoted execution.
- Equal distribution of load due to Hydraulic cylinder connected to the lift arm having two rods
- MOC-Stainless Steel Clad MS column / SS column.
- Adjustable mount stop
- Lifting all containers with forklift channel
- Precise positioning of IBC direction and height
- Double-hand operation





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## TECHNICAL SPECIFICATIONS

| MODEL                                  | SAD200                                                           | SAD250     | SAD300      | SAD400      |
|----------------------------------------|------------------------------------------------------------------|------------|-------------|-------------|
| DIAMETER (MM)                          | 200                                                              | 250        | 300         | 400         |
| VOLUME (LITRES)                        | 3                                                                | 6          | 12          | 20          |
| CAPACITY (DM <sup>3</sup> /HR)         | 0.5 - 1.2                                                        | 1.2 - 2.5  | 2.5 - 4.0   | 4.0 - 9.0   |
| FILTER                                 | P.T.F.E BAG FILTER OR STAINLESS STEEL AISI 316L SINTERED FILTER. |            |             |             |
| NUMBER OF FILTERS                      | 1                                                                | 1          | 1           | FROM 1 to 2 |
| INLET VALVE (BUTTERFLY VALVE)          | 2"                                                               | 2 - 1/2"   | 2 - 1/2"    | 3"          |
| VALVE ON VACUUM LINE (BUTTERFLY VALVE) | 2"                                                               | 2 - 1/2"   | 2 - 1/2"    | 3"          |
| VALVE FOR C.I.P (DIAPHRAGM VALVE)      | 3/4"                                                             | 3/4"       | 3/4"        | 3/4"        |
| OVERPRESSURE VALVE (DIAPHRAGM VALVE)   | 3/4"                                                             | 3/4"       | 3/4"        | 3/4"        |
| BOTTOM VALVE (BUTTERFLY VALVE) (MM)    | 150                                                              | 200        | 250         | 250         |
| CONNECTION WITH BOTTOM VALVE           | TRI CLAMP / FLANGE                                               |            |             |             |
| HEIGHT (MM)                            | 1130                                                             | 1390       | 1450        | 1770        |
| AIR CONSUMPTION (NI/MIN) AT 6 BARG     | 500 - 1000                                                       | 500 - 1000 | 1000 - 2000 | 2000 - 4000 |
| GASKETS                                | SILICONE / PTFE / EPDM                                           |            |             |             |



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